Ira A. Fulton Schools of Engineering
Arizona State University

CSE 579: Knowledge Representation and Reasoning

(Optional)Building an OWL Ontology

Purpose

This assignment requires you to complete an introductory-level tutorial aimed at newcomers to ontologies and those using formalisms other than OWL. It will give you hands-on experience in the construction of ontologies as you follow the instructions in the tutorial and build an OWL ontology using the domain of pizzas.

Objectives

Learners will be able to:

- Demonstrate Competence in Constructing OWL Ontologies
- Accurately Document and Submit Assignment Artifacts

Technology Requirements

Protege

Project Description

The tutorial is organized into three parts and comprises four specific exercises:

- Exercise 1: Create a Primitive Class Hierarchy (expected, missing some classes)
- Exercise 2: Add disjoints
- Exercise 3: Properties and Restrictions
- Exercise 4: Define a MeatyPizza

Please follow the overall steps listed here to successfully complete the assignment.

Directions

1. Download the assignment guide (see below), preview the entire tutorial and assigned tasks, and review the evaluation criteria section below.

a. CSE579_Unit-7_GradedAssignmentGuide_inc-tutorial.pdf

2. Access Protege on your local machine.
    a. You must use Protege installed on your local machine to complete this assignment.
3. Complete the tutorial and integrated problems.
    a. With the assignment guide PDF close at hand, complete the entire tutorial, saving the information indicated for submission on Coursera. You must save the following information and files:
        i. Your answer to Q1 (see page 12 in the guide)
        ii. Your screen shot of DL queries (see page 25 in the guide)
        iii. Your OWL file (see page 25 in the guide)
4. Prepare your items for submission.
    a. Save your OWL file file and prepare your screen shot file for upload using the following naming conventions:
    b. OWL file: FirstnameLastname_OWL_U7_mm-dd--yy
    c. Screen shot: FirstnameLastname_DL_U7_mm-dd--yy
    d. NOTE: Save your screen shot in one of the following formats: .png, .jpg, or .pdf

Submission Directions for Project Deliverables

You are given unlimited attempts to submit your best work. The number of attempts is given to anticipate any submission errors you may have in regards to properly submitting your best work within the deadline (e.g., accidentally submitting the wrong paper). It is not meant for you to receive multiple rounds of feedback and then one (1) final submission. Only your most recent submission will be assessed.

You must submit your Building an OWL Ontology deliverables in this submission space. Learners may not email or use other means to submit any assignment or project for review, including feedback, and grading.

The Building an OWL Ontology includes three (3) deliverables:

- **Answer to Quiz Question:** This assignment will have you complete an application quiz question.
- **Upload your DL Queries:** Use the file upload question to submit your DL queries

- **Upload your OWL file:** Use the file upload space to upload your OWL files.

To submit your work follow the directions below:

1. Navigate to your course and click on the "**Submission: (Optional)Building an OWL Ontology**" submission space
2. Click "**Take the Quiz**" in the upper right corner
3. Responsd to the questions.
4. There will be two submission spaces to upload your assignment documents.
5. When finished, click "**Submit Quiz**"
    a. If needed: to resubmit the assignment
        i. Click "**Take the Quiz Again**" and add your file submission again
        ii. When finished, click "**Submit Quiz**"

Evaluation

This assignment is staff-graded. An additional 2% may be added to your final grade after the successful submission of this assignment.

Rubric

Please review the rubric found on the submission page for this assignment.